

# After Simon, after Hrushovski

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ABSTRACT.

## 1. Patterns

In the following  $\mathcal{U}$  is some monster model of a complete theory  $T$  of language  $L$ . For simplicity  $T$  is assumed to be one-sorted.

We define a structure with domain  $\mathcal{U}^x$ , where  $x$  is some fixed tuple of variables. The language is denoted by  $L^P$ . Though  $L^P$  depends on  $x$ , this will be left implicit.

Let  $V$  be some large set of variables that are copies of  $x$ . Let  $\Delta$  be the set of positive Boolean combinations of formulas in  $\bigcup_{x' \in V} L_{x'; z}$ , where  $z = \langle z_i : i \in \omega \rangle$  is a fixed tuple of variables of length  $\omega$ . The choice of  $\Delta$  is justified in the remark below.

For every formula  $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$  and every type  $\alpha(z) \subseteq L$  the language  $L^P$  contains the  $n$ -ary relation  $r_{\varphi; \alpha}$ . We require that  $\alpha(z)$  is a finitary type, i.e. only a finite number of variables actually occur in it.

Below by the symbol  $L^P$  we will also denote the set of *atomic* formulas  $r_{\varphi; \alpha}(x_0, \dots, x_{n-1})$  for any  $x_0, \dots, x_{n-1} \in V$ . Note that equalities are *not* included in  $L^P$ .

For a given model weakly saturated model  $M$ , we define the interpretation of  $r_{\varphi; \alpha}$  in  $\mathcal{U}^x$ . Eventually, we will prove that the main definition in this section, the definition of *core*, does not depend on the choice of  $M$ . However, some of the intermediate notions introduced here do depend on  $M$ , though to avoid cluttering the notation we will omit reference to it.

If  $\bar{a} = a_0, \dots, a_{n-1} \in \mathcal{U}^x$ , we declare  $r_{\varphi; \alpha}(\bar{a})$  true if  $\varphi(\bar{a}; b)$  holds for every  $b \in M^z$  such that  $\alpha(b)$ . We repeat the definition for emphasis

$$r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow \forall z \in \alpha(M^z) \varphi(\bar{a}; z).$$

The following is paramount.

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**1 Remark** By the choice of  $\Delta$ , the truth of  $r_{\varphi;\alpha}(\bar{a})$  only depends on the equivalence class of  $\equiv_M$  of the components of the tuple  $\bar{a}$ . Explicitly, if  $\bar{a} = a_0, \dots, a_{n-1}$  and  $\bar{a}' = a'_0, \dots, a'_{n-1}$  are such that  $a_i \equiv_M a'_i$  for every  $i < n$ , then  $r_{\varphi;\alpha}(\bar{a}) \leftrightarrow r_{\varphi;\alpha}(\bar{a}')$ .

By the remark, it would be natural to define the interpretation of  $L^P$  on the quotient  $\mathcal{U}^x / \equiv_M$  or on the type space  $S_x(M)$ , but we prefer to keep the original domain  $\mathcal{U}^x$  to have a common playground with  $L$ .

A **p-type** is a set of (atomic) formulas in  $L^P(\mathcal{U}^x)$ .

Note incidentally that the relation  $x_0 \equiv_M x_1$  is definable by the p-type containing the formulas  $r_{\vartheta;\top}(x_0, x_1)$  for every  $\vartheta(x_0, x_1; z)$  of the form  $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$ .

We write **p-tp**( $\bar{a}$ ) for the p-type of  $\bar{a}$ , i.e. the set of formulas  $r_{\varphi;\alpha}(\bar{x})$  such that  $r_{\varphi;\alpha}(\bar{a})$  holds.

**2 Fact** Formulas in  $L^P$  are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

**Proof.** The claim about dummy variables is clear, so we tacitly use it below.

Up to renaming of the variables that occur in  $\beta(z)$  and  $\psi(\bar{x}; z)$ , we can assume that they are disjoint from those in  $\alpha(z)$ . Therefore the following equivalences hold

$$\begin{aligned} r_{\varphi;\alpha}(\bar{a}) \vee r_{\psi;\beta}(\bar{a}) &\Leftrightarrow \forall z \in \alpha(M^z) \varphi(\bar{a}; z) \vee \forall z \in \beta(M^z) \psi(\bar{a}; z) \\ &\Leftrightarrow \forall z \in \alpha(M^z) \cup \beta(M^z) \varphi(\bar{a}; z) \vee \psi(\bar{a}; z) \\ &\Leftrightarrow r_{\varphi \vee \psi; \alpha \cup \beta}(\bar{a}) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula  $\exists y r_{\varphi;\alpha}(\bar{x}, y)$ . Without loss of generality, we can assume that  $\varphi$  has the form

$$\varphi(\bar{x}, y; z) = \bigvee_{i=1}^m \varphi_i(\bar{x}, y; z)$$

where  $\varphi_i(\bar{x}, y; z) = \varphi'_i(\bar{x}; z) \wedge \psi_i(y; z)$  and each  $\varphi'_i(\bar{x}; z)$  is a conjunction of formulas in  $\bigcup_{x' \in V} L_{x'; z}$ . Closure under existential quantification follows from the following equivalences

$$\begin{aligned} \exists y r_{\varphi;\alpha}(\bar{a}, y) &\Leftrightarrow \exists y \forall z \left[ z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi_i(\bar{a}, y; z) \right] \\ &\Leftrightarrow \forall z \left[ z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi'_i(\bar{a}; z) \wedge \exists y \psi_i(y; z) \right] \\ &\Leftrightarrow r_{\vartheta;\alpha}(\bar{a}), \end{aligned}$$

where

$$\vartheta(\bar{x}; z) = \bigwedge_{i=1}^m \varphi'_i(\bar{x}; z) \wedge \exists y \psi_i(y; z)$$

is a positive Boolean combination of formulas in  $\bigcup_{x' \in V} L_{x'; z}$ , as required.  $\square$

**3 Fact ( $\infty$ -p-saturation)** Let  $\mathcal{A} \subseteq \mathcal{U}^x$  have arbitrary cardinality. Let  $\bar{x}$  be a tuple of copies of  $x$  of length  $\leq |\mathcal{U}|$ . Then every p-type  $p(\bar{x}) \subseteq L^p(\mathcal{A})$  that is finitely consistent in  $\mathcal{U}^x$  is realized in  $\mathcal{U}^x$ .

**Proof.** Let  $A \subseteq \mathcal{U}^x$  be a small set containing a representative of every  $\equiv_M$  equivalence class that intersects  $\mathcal{A}$ . Replacing  $\mathcal{A}$  in  $p(\bar{x})$  by elements  $A$  we obtain an equivalent type  $p'(\bar{x}) \subseteq L^p(A)$ . Note that every formula  $r_{\varphi; \alpha}(\bar{x})$  there is a type  $q(\bar{x}) \subseteq L$  such that

$$\mathcal{U}^x \models r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow \mathcal{U} \models q(\bar{a})$$

Therefore the saturation of  $\mathcal{U}$  implies that  $p'(\bar{x})$  is realized in  $\mathcal{U}^x$ .  $\square$

## 2. Morphism

As the interpretation of  $L^p$  factors through the equivalence relation  $\equiv_M$ , p-morphisms could be introduced as (partial) maps of  $\mathcal{U}^x / \equiv_M$  into itself. However there are a few occasions when it is preferable to work with maps of  $\mathcal{U}^x$  into itself. Though the switching between the two perspectives is intuitively obvious, it requires a fair amount of notation.

Let  $R \subseteq \mathcal{U}^x \times \mathcal{U}^x$  be a binary relation. If  $\bar{a} = \langle a_i : i < \lambda \rangle$  and  $\bar{b} = \langle b_i : i < \lambda \rangle$  we write  $\langle \bar{a}, \bar{b} \rangle \in R$  to abbreviate:  $\langle a_i, b_i \rangle \in R$  for all  $i < \lambda$ . We say that  $R$  is a **p-morphism** if the following condition holds

$$\# \quad \mathcal{U}^x \models r_{\varphi; \alpha}(\bar{a}) \Rightarrow \mathcal{U}^x \models r_{\varphi; \alpha}(\bar{b}) \quad \text{for every } r_{\varphi; \alpha} \in L^p \text{ and every } \langle \bar{a}, \bar{b} \rangle \in R.$$

When the converse implication in  $\#$  also holds, i.e. when also  $R^{-1}$  is a p-morphism, we say that  $R$  is a **partial p-isomorphism**. When  $\mathcal{A} \subseteq \text{dom } R$ ,  $\mathcal{B} \subseteq \text{range } R$ , and  $R \cap (\mathcal{A} \times \mathcal{B})$  is a partial p-isomorphism, we say that  $R$  is an **p-isomorphism** between  $\mathcal{A}$  and  $\mathcal{B}$  or, when  $\mathcal{A} = \mathcal{B}$ , a **p-automorphism** of  $\mathcal{A}$ .

If  $R$  is a p-morphism, also  $\equiv_M \circ R \circ \equiv_M$  is a p-morphism. This we call the **full** p-morphism generated by  $R$ . We say that  $R$  and  $R'$  are **equivalent** if they generate the same full p-morphism.

**4 Remark** We can identify p-morphisms with maps  $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$ . In fact, to any such map we can associate the relation  $R = \bigcup_{a \in \mathcal{U}^x} ((a / \equiv_M) \times f(a / \equiv_M))$ .

Vice versa, to any p-morphisms  $R$  we can associate  $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$  that maps  $a / \equiv_M \mapsto b / \equiv_M$  for some/any  $b$  such that  $\langle a, b \rangle \in R$ . This is well-defined because if  $\langle a, b \rangle, \langle a', b' \rangle \in R$  and  $a \equiv_M a'$ , then we necessarily have  $b \equiv_M b'$ .

We find more convenient to work with relations.

We say that the p-morphism  $R$  is **reduced** if there is no  $R' \subset R$  that is equivalent to  $R$ . It is clear that if  $A, B \subseteq \mathcal{U}^x$  are any two sets of representative of  $\equiv_M$  then  $f = R \cap (A \times B)$  is a reduced morphism equivalent to  $R$ .

**5 Fact** Any reduced p-morphism is the graph of a (partial) map. Therefore every reduced p-isomorphism is a bijection.

**Proof.** Suppose not, then there are  $a, b, c$  such that  $\langle a, b \rangle, \langle a, b' \rangle \in R$  but  $b \neq b'$ . As  $R$  is reduced, then  $b \not\equiv_M b'$ . As noted above, this is not possible. The second claim follows immediately.  $\square$

**6 Fact** Every p-morphism extends to a p-morphism defined on the whole of  $\mathcal{U}^x$ . Also, every p-isomorphism extends to a p-automorphism of  $\mathcal{U}^x$ .

**Proof.** Let  $R$  be a p-morphism. Pick be a reduced p-morphism  $f$  equivalent to  $R$ . Let  $\bar{a}$  be an enumeration of  $\text{dom } f$  and let  $b \in \mathcal{U}^x$ . Let  $p(\bar{x}) = \text{p-tp}(\bar{a})$  and let  $q(\bar{x}, y) = \text{p-tp}(\bar{a}, b)$ . By  $\infty$ -p-saturation  $\exists y q(\bar{x}, y)$  is equivalent to the type containing the existential closure of the formulas in  $q(\bar{x}, y)$ . By Fact 2, the latter is equivalent to a p-type, which is necessarily coincides with  $p(\bar{x})$ . Then  $q(f\bar{a}, y)$  is consistent hence by  $\infty$ -p-saturation it is realized by some  $c \in \mathcal{U}^x$ . Then  $f \cup \{\langle b, c \rangle\}$  is a p-morphism defined in  $b$  that extends  $f$ . Then  $R$  can be extended as well.

The second claim is obtained by back-and-forth.  $\square$

### 3. Duality

Let  $m$  be an enumeration of  $M$ .

**7 Remark** We remark an simple fact which is used in the proof below. We say that  $\varphi'(a; z)$  is an alphabetic variant of  $\varphi(a; z)$  if it is obtained by replacing the variables effectively occurring in  $\varphi(a; z)$  by other variables (among those in  $z$ ). The following are clearly equivalent

1.  $\varphi(a; M^z) = M^z$
2.  $\varphi'(a; m)$  holds for every alphabetic variant of  $\varphi(a; z)$ .

Let  $S_{m,M} \subseteq S_z(\mathcal{U})$ , where  $z$  has the length of  $m$ , be the set of global types that extend  $\text{tp}(m/\emptyset)$  and are finitely satisfied in  $M$ .

**8 Fact** Assume  $x$  has infinite length. Then there is a one-to-one correspondence between the elements of  $S_{m,M}$  and the total p-morphisms of  $\mathcal{U}^x$  up to equivalence.

**Proof.** As  $x$  has infinite length, we can assume that the formulas in the types in  $S_z(\mathcal{U})$  are the form  $\varphi(a; z)$  for some  $\varphi(x; z) \in L$  and  $a \in \mathcal{U}^x$ .

Let  $R \subseteq \mathcal{U}^x \times \mathcal{U}^x$  be a total p-morphism. Let  $p(z)$  be the type containing the formulas  $\varphi(a; z)$  such that  $\varphi(b; m)$  for some/any  $b$  such that  $\langle a, b \rangle \in R$ . We claim that  $p(z) \in S_{m, M}$ . It easy to see that  $p(z)$  is complete. The inclusion  $p(z) \supseteq \text{tp}(m/\emptyset)$  is clear, so we only need to prove that  $p(z)$  is finitely satisfied in  $M$ . Suppose not, then  $\psi(a_1, \dots, a_n; M^z) = \emptyset$ , where

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

for some  $\varphi_i(x; z) \in L$  and  $\langle a_i, b_i \rangle \in R$  such that  $\varphi_i(b_i; m)$ . Then  $r_{\neg\psi; \top}(a_1, \dots, a_n)$  and  $\neg r_{\neg\psi; \top}(f a_1, \dots, f a_n)$ . A contradiction.

Vice versa, let  $p(z) \in S_{m, M}$  be given. Let  $R \subseteq \mathcal{U}^x \times \mathcal{U}^x$  contain the pairs  $\langle a, b \rangle$  such that  $\varphi(a; z) \in p \Leftrightarrow \varphi(b; m)$  holds for every  $\varphi(x; z) \in L$ . We prove that  $R$  is a p-morphism.

Let  $\psi(x_1, \dots, x_n; z) \in \Delta$ ,  $\alpha(z) \subseteq L$ , and  $a_1, \dots, a_n \in \mathcal{U}^x$  be such that  $r_{\psi; \alpha}(a_1, \dots, a_n)$ . Finally, let  $\langle a_i, b_i \rangle \in R$ . We can assume  $\alpha(M^z) \neq \emptyset$ , otherwise  $r_{\psi; \alpha}(b_1, \dots, b_n)$  trivially holds. Assume

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(x_{i_{h,k}}; z)$$

for some  $i_{h,k} \in \{1, \dots, n\}$ .

As  $r_{\psi; \alpha}(a_1, \dots, a_n)$ , up to re-indexing, we obtain that  $\varphi_{0,k}(a_{i_{0,k}}; c)$  every  $c \in \alpha(M^z)$ . As  $p(z)$  is finitely satisfiable in  $M$ , then  $\alpha(z) \wedge \neg \varphi_{0,k}(a_{i_k}; z) \notin p$ . Then, by completeness,  $p(z) \vdash \alpha(z) \rightarrow \varphi_{0,k}(b_{i_k}; z)$ . Then  $m \models \alpha(z) \rightarrow \varphi_{0,k}(b_{i_k}; z)$ .

□

#### 4. Cores

A set  $\mathcal{A} \subseteq \mathcal{U}^x$  is **p-maximal** if it satisfies the following (obviously) equivalent conditions for every  $\bar{a} \in \mathcal{A}^{\bar{x}}$ , where  $\bar{x}$  is a finite tuple, and  $\bar{b}$  such that  $\langle \bar{a}, \bar{b} \rangle \in R$  for some p-morphism  $R$

- a.  $r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow r_{\varphi; \alpha}(\bar{b})$  for every  $r_{\varphi; \alpha} \in L^p$
- b.  $\text{p-tp}(\bar{a}) = \text{p-tp}(\bar{b})$

**9 Fact** The following are equivalent

1.  $\mathcal{A} \subseteq \mathcal{U}^x$  is p-maximal
2.  $p(\bar{x}) = \text{p-tp}(\bar{a})$  is a maximally consistent p-type for every  $\bar{a} \in \mathcal{A}^{\bar{x}}$  and every finite  $\bar{x}$
3.  $p(\bar{y}) = \text{p-tp}(\bar{c})$  is a maximally consistent p-type for  $\bar{c}$  any enumeration of  $\mathcal{A}$
4. every p-morphism defined on  $\mathcal{A}$  is a p-isomorphism between  $\mathcal{A}$  and its image.

**Proof.** 1 $\Rightarrow$ 2. Negate 2. Suppose  $\text{p-tp}(\bar{a})$  is not maximal for some finite  $\bar{a} \in \mathcal{A}^{\bar{x}}$ . Pick some  $\bar{a}' \in \mathcal{A}^{\bar{x}}$  such that  $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$ . Then  $R = \{\langle \bar{a}, \bar{a}' \rangle\}$  is a p-morphism that contradicts the p-maximality of  $\mathcal{A}$ .

2 $\Rightarrow$ 3. If  $\bar{c}$  is an enumeration of  $\mathcal{A}$  that witnesses the failure of 3, then some finite subtuple of  $\bar{c}$  witnesses the failure of 2.

3 $\Rightarrow$ 4. Negate 4. Let  $R$  be a p-morphism defined on  $\mathcal{A}$  that is not a p-isomorphism. Let  $\bar{a} \in \mathcal{A}^{\bar{x}}$  and  $\bar{b}$  be such that  $\langle \bar{a}, \bar{b} \rangle \in R$  and  $\text{p-tp}(\bar{a}) \setminus \text{p-tp}(\bar{b}) \neq \emptyset$ . To contradict 3, it suffices to extend  $\{\langle \bar{a}, \bar{b} \rangle\}$  to a p-morphism defined on  $\mathcal{A}$ . This is possible by  $\infty$ -p-homogeneity.

4 $\Rightarrow$ 1. Clear.  $\square$

**10 Fact** If  $\mathcal{A}$  is p-maximal, also  $(\equiv_M)[\mathcal{A}] = \{a : \exists x \ a \equiv_M x \in \mathcal{A}\}$  is p-maximal.

**Proof.** By 4 in Fact 9. Indeed, let  $R$  be a p-morphism defined on  $(\equiv_M)[\mathcal{A}]$ . As  $R \circ (\equiv_M)$  is also a p-morphism,  $R \circ (\equiv_M)[\mathcal{A}]$  is p-maximal.  $\square$

**11 Example** Assume that  $a \in \mathcal{U}^x$  is such that  $\varphi(a; M^z)$  is definable over  $\text{acl}^{\text{eq}} \emptyset$  for every  $\varphi(x; z) \in L$ . Then the orbit of  $a$  under  $\text{Aut}(\mathcal{U}/\{M\})$  is p-maximal.

**Proof.** First consider the case when  $\varphi(a; M^z)$  is definable over  $\emptyset$ . We prove that  $\{a\}$  is p-maximal.

Let  $\varphi'(z) \in L$  be such that  $\varphi(a; M^z) \leftrightarrow \varphi'(M^z)$ . Then, if we set  $\varphi''(x; z) = \varphi(x; z) \leftrightarrow \varphi'(z)$ , the formula  $r_{\varphi'', \top}(a)$  holds.

If  $\{\langle a, b \rangle\}$  is a p-morphism then  $r_{\varphi'', \top}(b)$  also holds. Then  $a \equiv_M b$  follows. This shows that p-morphisms are trivial and therefore p-isomorphisms.

Now assume that  $\varphi(a; M^z)$  is definable over  $\text{acl}^{\text{eq}} \emptyset$ .  $\square$

A set  $\mathcal{J} \subseteq \mathcal{U}^x$  is called a **core** if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

**12 Fact** Let  $\mathcal{J}$  be a core. Let  $\mathcal{A}$  be a p-maximal set. Then every p-morphism  $R \subseteq \mathcal{A} \times \mathcal{J}$  extends to a p-morphism  $R'$  such that  $\mathcal{A} = \text{dom} R'$  and  $\text{range} R' \subseteq \mathcal{J}$ .

**Proof.** Reason as in the proof of Fact 6 with the same notation but choose  $b \in \mathcal{A}$ . Let  $R'$  be a p-morphism defined on  $\mathcal{A}$  that extends  $f \cup \{\langle b, c \rangle\}$ . Then  $R'[\mathcal{A}]$ , i.e. the image of  $\mathcal{A}$  under  $R'$ , is a p-maximal set that contains  $\mathcal{J}$ . By maximality  $R'[\mathcal{A}]$  coincides with  $\mathcal{J}$ , hence  $c \in \mathcal{J}$ .  $\square$

By back-and-forth we obtain the following.

**13 Corollary ( $\infty$ -p-homogeneity of the core)** If  $\mathcal{J}$  and  $\mathcal{J}'$  are cores, then every p-morphism  $R \subseteq \mathcal{J} \times \mathcal{J}'$  extends to a p-isomorphism between  $\mathcal{J}$  and  $\mathcal{J}'$ .

A **p-retraction** of  $\mathcal{U}^x$  onto  $\mathcal{A}$  is a total p-morphism  $R$  that coincides with  $\equiv_M$  on  $\mathcal{A}$ , i.e. such that  $R \cap (\mathcal{A} \times \mathcal{A}) = (\equiv_M) \cap (\mathcal{A} \times \mathcal{A})$ .

**14 Theorem** If  $\mathcal{J}$  is a core then there is a p-retraction of  $\mathcal{U}^x$  onto  $\mathcal{J}$

**Proof.** Let  $\bar{c}$  enumerate  $\mathcal{U}^x$ . Let  $\bar{a}$  be a tuple such that  $\text{p-tp}(\bar{a}) \supseteq \text{p-tp}(\bar{c})$  is p-maximal. By Fact 10 the relation  $R = (\equiv_M \circ \{\langle \bar{c}, \bar{a} \rangle\} \circ \equiv_M)$  is a p-morphism onto the p-maximal set  $\mathcal{A} = \text{range } R$ . As  $\mathcal{A}$  is p-maximal, by Fact 12 we can extend  $(R \cap (\mathcal{J} \times \mathcal{A}))^{-1}$  to a p-isomorphism  $R'$  defined on  $\mathcal{A}$ . It is easy to verify that  $R' \circ R$  is a retraction.  $\square$

**15 Fact** The following are equivalent

1.  $a \in \bigcap \{\mathcal{J} : \mathcal{J} \text{ is a core}\}$
2.  $\varphi(a; M^z)$  is definable over  $\text{acl}^{\text{eq}} \emptyset$ , for every formula  $\varphi(x; z) \in L$ .

**Proof.** ???  $\square$

**16 Definition** Let  $\mathcal{J}$  be a core; we define a topology on  $\mathcal{J}$ . The closed sets in the p-topology on  $\mathcal{J}$  are those defined by p-types with (arbitrary many) parameters in  $\mathcal{J}$ .

**17 Fact** The p-topology is compact and  $T_1$ .

**Proof.** ???  $\square$